

WI4014TU – Numerical Analysis for PDE's

Lecture 2

Classification and Theoretical Analysis of PDE's

Classification of PDE's

Second-order PDE's with two variables

The second-order partial differential equation with two independent variables x and y is a relation between the unknown function $u(x, y)$ and its partial derivatives up to and including the second-order:

$$F\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial x \partial y}, \frac{\partial^2 u}{\partial y^2}\right) = 0$$

Linear second-order PDE's

A second-order PDE is **linear with respect to the highest-order derivatives** if it has the form:

$$a_{11} \frac{\partial^2 u}{\partial x^2} + 2a_{12} \frac{\partial^2 u}{\partial x \partial y} + a_{22} \frac{\partial^2 u}{\partial y^2} + F_1 \left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = 0$$

where the coefficients a_{11} , a_{12} , and a_{22} are functions of x and y only. The equation is **quasi-linear** if these coefficients are also functions of u , $\frac{\partial u}{\partial x}$, and $\frac{\partial u}{\partial y}$. The equation is **linear** if it has the form:

$$a_{11} \frac{\partial^2 u}{\partial x^2} + 2a_{12} \frac{\partial^2 u}{\partial x \partial y} + a_{22} \frac{\partial^2 u}{\partial y^2} + b_1 \frac{\partial u}{\partial x} + b_2 \frac{\partial u}{\partial y} + cu + f = 0$$

and a_{11} , a_{12} , a_{22} , b_1 , b_2 , c , f are functions of x and y only. If these functions are constants, then the equation is **linear with constant coefficients**.

Matrix of coefficients

Consider the linear (in highest derivatives) case:

$$a_{11} \frac{\partial^2 u}{\partial x^2} + 2a_{12} \frac{\partial^2 u}{\partial x \partial y} + a_{22} \frac{\partial^2 u}{\partial y^2} + F \left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = 0$$

This we can also write as

$$\begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} + F = 0$$

or simply as

$$\nabla \cdot \mathbb{A} \nabla u + F = 0,$$

where $\mathbb{A}$ is the matrix of coefficients.

Change of variables

Since $\mathbb{A}$ is symmetric, $\mathbb{A} = \mathbb{Q}\Lambda\mathbb{Q}^T$ and

$$\nabla \cdot \mathbb{Q}\Lambda\mathbb{Q}^T \nabla u + F = 0,$$

$$\tilde{\nabla} \cdot \Lambda \tilde{\nabla} u + F = 0,$$

where Λ is the diagonal matrix of eigenvalues of $\mathbb{A}$ and $\tilde{\nabla}$ is the gradient operator with respect to new variables ξ and η . The PDE becomes

$$\lambda_1 \frac{\partial^2 u}{\partial \xi^2} + \lambda_2 \frac{\partial^2 u}{\partial \eta^2} + F\left(\xi, \eta, u, \frac{\partial u}{\partial \xi}, \frac{\partial u}{\partial \eta}\right) = 0$$

Hint, $u(x(\xi, \eta), y(\xi, \eta))$:

$$\frac{\partial u}{\partial \xi} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial \xi} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial \xi} = \frac{\partial u}{\partial x} q_{11} + \frac{\partial u}{\partial y} q_{12}$$

Classification of second-order PDE's

Eigenvalues λ_1 and λ_2 of the symmetric $\mathbb{A}$ are real. There are three distinct possibilities:

- ① λ_1 and λ_2 both nonzero and have the same sign. Equation is **elliptic**, e.g.:

$$\frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \eta^2} + F\left(\xi, \eta, u, \frac{\partial u}{\partial \xi}, \frac{\partial u}{\partial \eta}\right) = 0$$

- ② λ_1 and λ_2 both nonzero and have different signs. Equation is **hyperbolic**, e.g.:

$$\frac{\partial^2 u}{\partial \xi^2} - \frac{\partial^2 u}{\partial \eta^2} + F\left(\xi, \eta, u, \frac{\partial u}{\partial \xi}, \frac{\partial u}{\partial \eta}\right) = 0$$

- ③ One of λ_1 and λ_2 is zero. Equation is **parabolic**, e.g.:

$$\frac{\partial^2 u}{\partial \xi^2} + F\left(\xi, \eta, u, \frac{\partial u}{\partial \xi}, \frac{\partial u}{\partial \eta}\right) = 0$$

How to classify a second-order PDE

Reduce the original PDE to the form

$$a_{11} \frac{\partial^2 u}{\partial x^2} + 2a_{12} \frac{\partial^2 u}{\partial x \partial y} + a_{22} \frac{\partial^2 u}{\partial y^2} + F \left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = 0$$

Expressing the eigenvalues of the coefficient matrix $\mathbb{A}$ via the coefficients a_{11} , a_{12} , and a_{22} , we find:

- ① Equation is **elliptic** if $a_{12}^2 - a_{11}a_{22} < 0$
- ② Equation is **hyperbolic** if $a_{12}^2 - a_{11}a_{22} > 0$
- ③ Equation is **parabolic** if $a_{12}^2 - a_{11}a_{22} = 0$

The class of PDE may change throughout time and space, since coefficients may depend on t , x , and y .

Example: Poisson's equation

Poisson's equation with differentiable $k(x, y)$ is elliptic:

$$-\nabla \cdot (k \nabla u) = f$$

$$-k \nabla \cdot \nabla u - (\nabla k) \cdot (\nabla u) = f$$

$$-k \frac{\partial^2 u}{\partial x^2} - k \frac{\partial^2 u}{\partial y^2} - \frac{\partial k}{\partial x} \frac{\partial u}{\partial x} - \frac{\partial k}{\partial y} \frac{\partial u}{\partial y} = f$$

Two variables x and y . Usually, $k(x, y) > 0$.

Poisson's and Laplace's equations are elliptic in three dimensions as well: signs in front of all three second-order derivatives are the same.

Example: wave equation

Wave equation:

$$\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f$$

There are three variables in $u(x, y, t)$: x , y , and t . Same eigenvalue analysis as in $2D$ but with a 3×3 matrix of coefficients. Similar conclusions about the signs of coefficients. Coefficients in front of the second-order spatial derivatives have different sign compared to the second-order time derivative. Hence, wave equation is hyperbolic.

Wave equation is hyperbolic in three dimensions as well:

$$\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} - \Delta u = f$$

Four variables: x , y , z , and t . Sign in front of the second-order derivative with respect to time t is different from the rest.

Example: advection equation

The advection equation is of the first-order:

$$\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = f$$

but is also considered to be hyperbolic. Similar to the wave equation.

Example: diffusion equation

Diffusion equation:

$$\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} = f$$

There are two variables x and t . Coefficient in front of the second-order derivative with respect to t is zero. Hence, diffusion equation is parabolic.

Diffusion equation is parabolic in two and three dimensions as well:

$$\frac{\partial u}{\partial t} - \nabla \cdot k \nabla u = f$$

Four variables: x , y , z , and t . Second derivative in t is missing.

Example: convection-diffusion equation

Convection-diffusion equation

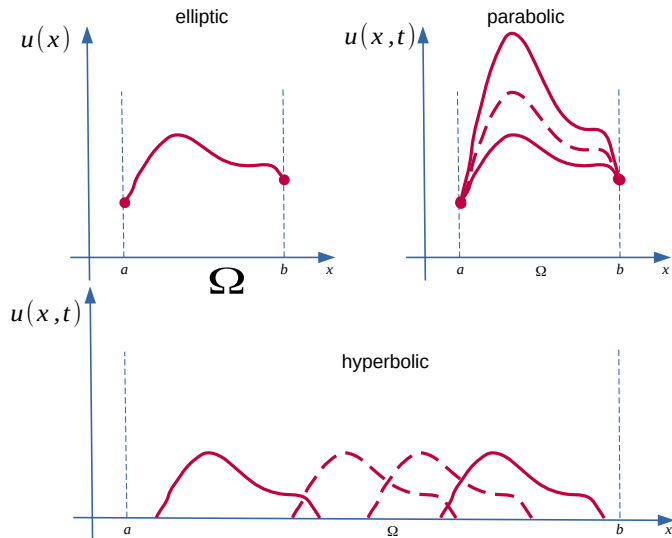
$$\frac{\partial u}{\partial t} - \nabla \cdot k \nabla u + \mathbf{v} \cdot \nabla u = f$$

is formally parabolic if $k > 0$. However, if $k \ll |\mathbf{v}|$, then convection *dominates* over diffusion, and the equation is almost hyperbolic. This is called a **singular perturbation** of the problem.

Why do we classify equations?

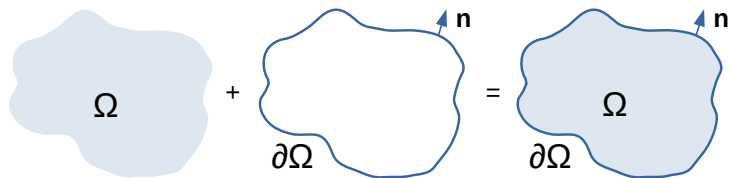
- Elliptic equations describe steady-state problems
- Parabolic equations describe very smooth dynamics (diffusion) towards steady state
- Hyperbolic equations describe waves, i.e., their solutions move through space, often preserving their shape

Typical behavior



Boundary and initial conditions

Domain Ω , boundary surface $\partial\Omega$, normal vector $\mathbf{n}$



$$\Omega \cup \partial\Omega = \bar{\Omega}$$

Boundary conditions for Poisson's equation

Let $u(\mathbf{x})$ satisfy

$$-\nabla \cdot k \nabla u = f$$

for all $\mathbf{x} \in \Omega \subset \mathbb{R}^p$, $p = 1, 2, 3$. Let $\partial\Omega$ be the boundary of Ω with the outwards normal vector $\mathbf{n}$. The function u may be required to satisfy the following common boundary conditions (BC):

- ❶ **Dirichlet BC:** $u(\mathbf{x}) = g_0(\mathbf{x})$, $\mathbf{x} \in \partial\Omega$
- ❷ **Neumann BC:** $k(\mathbf{x})D_{\mathbf{n}}u(\mathbf{x}) = g_1(\mathbf{x})$, $\mathbf{x} \in \partial\Omega$
- ❸ **Robin (mixed) BC:** $k(\mathbf{x})D_{\mathbf{n}}u(\mathbf{x}) + \sigma u(\mathbf{x}) = g_2(\mathbf{x})$, $\mathbf{x} \in \partial\Omega$

where $D_{\mathbf{n}}u$, also denoted as $\partial u / \partial \mathbf{n}$ is the directional derivative of u , and $g_{0,1,2}(\mathbf{x})$ are given functions.

The boundary $\partial\Omega$ may be split into several disjoint parts. And u may be required to satisfy different BC's at different parts of $\partial\Omega$.

Example: electrostatic potential

The electrostatic potential ϕ is maintained at the value of 5 V at the metallic piece $\partial\Omega_1$ of the boundary $\partial\Omega$ surrounding the domain Ω . The normal component of the electric field $\mathbf{E} = -\nabla\phi$ is zero at the rest of the boundary $\partial\Omega_2$. The **boundary value problem** for ϕ is:

$$\begin{aligned} -\Delta\phi(\mathbf{x}) &= 0, & \mathbf{x} \in \Omega \\ \phi(\mathbf{x}) &= 5, & \mathbf{x} \in \partial\Omega_1 \\ \mathbf{n} \cdot \nabla\phi(\mathbf{x}) &= D_n\phi(\mathbf{x}) = 0, & \mathbf{x} \in \partial\Omega_2 \end{aligned}$$

Boundary and initial conditions for diffusion equation

Let $u(\mathbf{x}, t)$ satisfy

$$\frac{\partial u}{\partial t} - \nabla \cdot k \nabla u = f$$

for all $\mathbf{x} \in \Omega \subset \mathbb{R}^p$, $p = 1, 2, 3$ and all $t \geq 0$. The following BC's are common:

- ❶ Dirichlet BC: $u(\mathbf{x}, t) = g_0(\mathbf{x}, t)$, $\mathbf{x} \in \partial\Omega$, $t \geq 0$
- ❷ Neumann BC: $k(\mathbf{x}) D_n u(\mathbf{x}, t) = g_1(\mathbf{x}, t)$, $\mathbf{x} \in \partial\Omega$, $t \geq 0$
- ❸ Robin BC: $k(\mathbf{x}) D_n u(\mathbf{x}, t) + \sigma u(\mathbf{x}, t) = g_2(\mathbf{x}, t)$, $\mathbf{x} \in \partial\Omega$, $t \geq 0$
- ❹ **Initial condition:** $u(\mathbf{x}, t) = u_0(\mathbf{x})$ at $t = 0$

where $g_{0,1,2}(\mathbf{x}, t)$ and $u_0(\mathbf{x})$ are given functions.

Example: heat flow

Find the temperature distribution $T(\mathbf{x}, t)$ inside a metallic rod with the thermal conductivity k . Initial temperature of the rod is T_1 . The temperature at one end of the rod remains T_1 at all times. The heat flux density through the other end of the rod is $q(t)$, $t \geq 0$. The rest of the rod surface exchanges heat with the environment at temperature T_0 by thermal radiation.

$$\frac{\partial T}{\partial t} - k\Delta T = 0, \quad \mathbf{x} \in \Omega, \quad t > 0$$

$$T(\mathbf{x}, 0) = T_1, \quad \mathbf{x} \in \Omega$$

$$T(\mathbf{x}, t) = T_1, \quad \mathbf{x} \in \partial\Omega_1, \quad t \geq 0$$

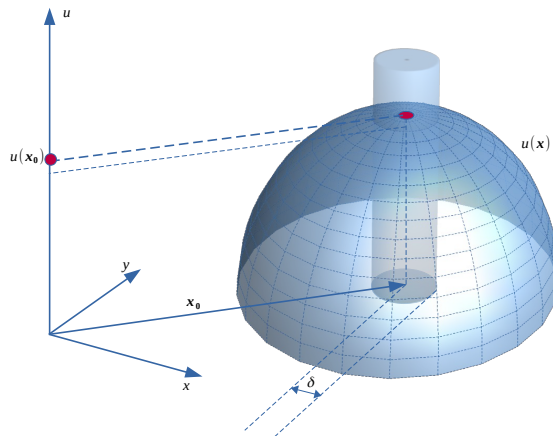
$$kD_{\mathbf{n}}T(\mathbf{x}, t) = -q(t), \quad \mathbf{x} \in \partial\Omega_2, \quad t \geq 0$$

$$kD_{\mathbf{n}}T(\mathbf{x}, t) + \lambda(T(\mathbf{x}, t) - T_0) = 0, \quad \mathbf{x} \in \partial\Omega_3, \quad t \geq 0$$

where $\partial\Omega_1$ is the surface with prescribed temperature, $\partial\Omega_2$ is the surface with prescribed flux density, and $\partial\Omega_3$ is the surface where the heat is exchanged with the environment.

Maximum Principle and Uniqueness of Solutions

Maximum of a function



What happens at and around a maximum

Definition (maximum): A function $u(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^2$ has a maximum at $\mathbf{x}_0 \in \Omega \subset \mathbb{R}^2$ if there exists $\delta > 0$ such that $u(\mathbf{x}_0) \geq u(\mathbf{x})$ for all $\mathbf{x}$, $\|\mathbf{x} - \mathbf{x}_0\| < \delta$.

Definition (Hessian): The hessian matrix of $u(\mathbf{x})$ at $\mathbf{x}_0 \in \mathbb{R}^2$ is defined as

$$H(\mathbf{x}_0) = \begin{bmatrix} \frac{\partial^2 u}{\partial x^2}(\mathbf{x}_0) & \frac{\partial^2 u}{\partial x \partial y}(\mathbf{x}_0) \\ \frac{\partial^2 u}{\partial y \partial x}(\mathbf{x}_0) & \frac{\partial^2 u}{\partial y^2}(\mathbf{x}_0) \end{bmatrix}$$

Necessary condition: For sufficiently smooth functions, such as $u(\mathbf{x}) \in C^2(\Omega)$, if $u(\mathbf{x})$ has a maximum at $\mathbf{x}_0$, then $\nabla u(\mathbf{x}_0) = \mathbf{0}$.

Theorem: If a function $u(\mathbf{x}) \in C^2(\Omega)$, $\Omega \subset \mathbb{R}^2$ has a maximum at $\mathbf{x}_0$, then there exists $\delta > 0$ such that for all $\mathbf{x}$, $\|\mathbf{x} - \mathbf{x}_0\| < \delta$, the Hessian matrix of $u(\mathbf{x})$ at $\mathbf{x}_0$ is negative semi-definite, i.e.

$(\mathbf{x} - \mathbf{x}_0)^T H(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) \leq 0$, for all $\mathbf{x}$, $\|\mathbf{x} - \mathbf{x}_0\| < \delta$

From Hessian to Laplacian

Observation: If $H(\mathbf{x}_0)$ is negative semi-definite, then $-\Delta u(\mathbf{x}_0) \geq 0$.

Corollary: Assume that $u(\mathbf{x})$ is defined and sufficiently smooth in the neighborhood of $\mathbf{x}_0$. Then,

- If $u(\mathbf{x})$ has a maximum in $\mathbf{x}_0$, then $\nabla u(\mathbf{x}_0) = \mathbf{0}$ and $-\Delta u(\mathbf{x}_0) \geq 0$
- If $u(\mathbf{x})$ has a minimum in $\mathbf{x}_0$, then $\nabla u(\mathbf{x}_0) = \mathbf{0}$ and $-\Delta u(\mathbf{x}_0) \leq 0$

Harmonic functions

Definition: A function $u(\mathbf{x})$ satisfying:

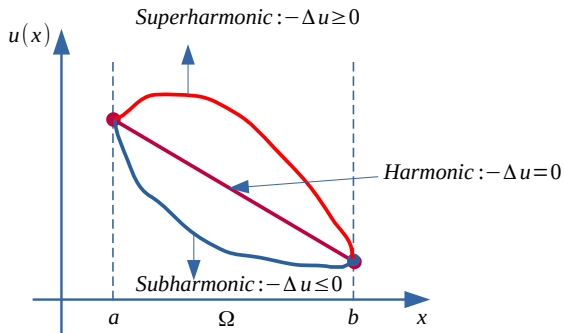
- $-\Delta u = 0$ for $\mathbf{x} \in \Omega$ is called **harmonic** on Ω
- $-\Delta u \leq 0$ for $\mathbf{x} \in \Omega$ is called **subharmonic** on Ω
- $-\Delta u \geq 0$ for $\mathbf{x} \in \Omega$ is called **superharmonic** on Ω

Maximum Principle

Theorem (Maximum Principle): Let Ω be an open bounded domain with boundary $\partial\Omega$ and closure $\overline{\Omega} = \Omega \cup \partial\Omega$. Suppose that $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ is *subharmonic* on Ω . Then,

- 1 $u(\mathbf{x})$ takes its maximum at the boundary $\partial\Omega$, hence,
$$\max_{\mathbf{x} \in \overline{\Omega}} u = \max_{\mathbf{x} \in \partial\Omega} u$$
- 2 Furthermore, if Ω is connected and if there is an internal point $\mathbf{x}_0$ where $u(\mathbf{x})$ reaches its maximum $u(\mathbf{x}_0) = \max_{\mathbf{x} \in \overline{\Omega}} u$, then $u(\mathbf{x})$ is constant on $\overline{\Omega}$, i.e., $u(\mathbf{x}) = u(\mathbf{x}_0)$ on $\overline{\Omega}$.

A similar (Minimum Principle) theorem holds for *superharmonic* functions with maximum replaced by the minimum.



Uniqueness of the solution with Dirichlet BC's

Theorem: Laplace's equation $-\Delta u = 0$ on Ω with a *homogeneous* Dirichlet boundary condition $u(\mathbf{x}) = 0$, $\mathbf{x} \in \partial\Omega$, has only the trivial solution $u(\mathbf{x}) = 0$ on Ω .

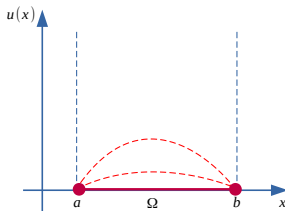
Theorem (uniqueness, Dirichlet): Let $\Omega \subset \mathbb{R}^2$ be a bounded region with boundary $\partial\Omega$, and suppose that $u \in C^2(\Omega) \cap C^1(\partial\Omega)$ satisfies

$$\begin{aligned} -\Delta u &= f(\mathbf{x}), & \mathbf{x} \in \Omega \\ u &= g(\mathbf{x}), & \mathbf{x} \in \partial\Omega \end{aligned}$$

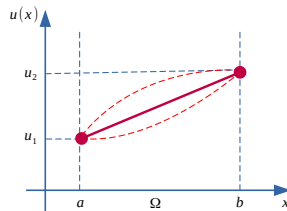
Then there exists at most one such solution $u(\mathbf{x})$.

Uniqueness of the solution with Dirichlet BC's

$$-u''=0, u(a)=0, u(b)=0$$



$$-u''=0, u(a)=u_1, u(b)=u_2$$



Uniqueness of the solution with Robin BC's

Theorem (uniqueness, Robin): Let $\Omega \subset \mathbb{R}^2$ be a bounded region with boundary $\partial\Omega$, and suppose that $u \in C^2(\Omega) \cap C^1(\partial\Omega)$ satisfies

$$\begin{aligned} -\Delta u &= f(\mathbf{x}), & \mathbf{x} \in \Omega \\ \sigma u + D_{\mathbf{n}}u &= g(\mathbf{x}), & \mathbf{x} \in \partial\Omega \end{aligned}$$

with $\sigma > 0$. Then there exists at most one such solution $u(\mathbf{x})$.

Non-uniqueness of the solution with Neumann BC's

Theorem (non-uniqueness, Neumann): Let $\Omega \subset \mathbb{R}^2$ be a bounded region with boundary $\partial\Omega$, and suppose that $u \in C^2(\Omega) \cap C^1(\partial\Omega)$ satisfies

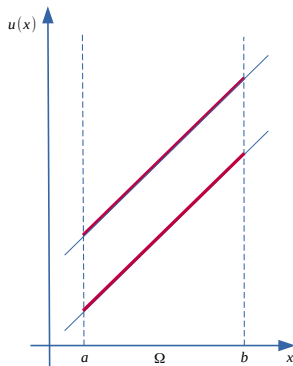
$$\begin{aligned} -\nabla \cdot k \nabla u &= f(\mathbf{x}), & \mathbf{x} \in \Omega \\ -k D_{\mathbf{n}} u &= g(\mathbf{x}), & \mathbf{x} \in \partial\Omega \end{aligned}$$

Then the following *compatibility condition* must be satisfied:

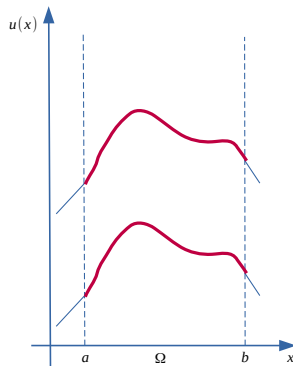
$$\int_{\partial\Omega} g \, ds = \iint_{\Omega} f \, dA$$

and the solution $u(\mathbf{x})$ is determined up to an additive constant only, i.e., $u(\mathbf{x}) + c$ is also a solution.

Non-uniqueness of the solution with Neumann BC's



$$-u''=0, u'(a)=1, -u'(b)=-1$$



$$-u''=f, u'(a)=g_1, -u'(b)=g_2$$

Non-negativity

Theorem (non-negativity): Let $\Omega \subset \mathbb{R}^2$ be a bounded region with boundary $\partial\Omega$, and suppose that $u \in C^2(\Omega) \cap C^1(\partial\Omega)$ satisfies

$$\begin{aligned} -\Delta u &\geq 0, & \mathbf{x} \in \Omega \\ u &= 0, & \mathbf{x} \in \partial\Omega \end{aligned}$$

Then $u(\mathbf{x}) \geq 0$ on Ω .

- This allows figuring out the sign of u , if $-\Delta u = f$, $\mathbf{x} \in \Omega$; $u = 0$, $\mathbf{x} \in \partial\Omega$, and the sign of f is given.
- What if $-\Delta u = f$, $\mathbf{x} \in \Omega$; $u = g$, $\mathbf{x} \in \partial\Omega$, and the signs of f and g are given?

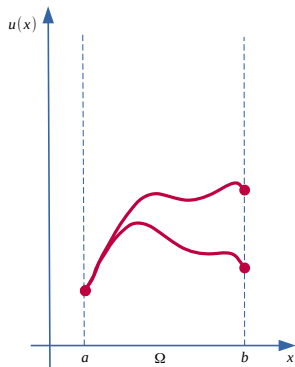
Stability with respect to changes in BC

Theorem (stability to changes in BC): Let $\Omega \subset \mathbb{R}^2$ be a bounded region with boundary $\partial\Omega$, and suppose that $u_1, u_2 \in C^2(\Omega) \cap C^1(\partial\Omega)$ satisfy

$$\begin{cases} -\Delta u_1 = f, & \mathbf{x} \in \Omega; \\ u_1 = g_1, & \mathbf{x} \in \partial\Omega; \end{cases}$$
$$\begin{cases} -\Delta u_2 = f, & \mathbf{x} \in \Omega; \\ u_2 = g_2, & \mathbf{x} \in \partial\Omega; \end{cases}$$

Then, $|u_1(\mathbf{x}) - u_2(\mathbf{x})| \leq \max_{\mathbf{x} \in \partial\Omega} |g_1(\mathbf{x}) - g_2(\mathbf{x})|$, for all $\mathbf{x} \in \Omega$.

Stability of the solution



$$-u''=f, u(a)=g_1, u(b)=g_2 \rightarrow u(b)=g_3$$